

SIMULATORS

Some systems are too complex for even supercomputers to simulate—particularly systems with properties that cannot be classically simulated, such as entanglement. However, these can be simulated with analog systems of real particles with quantum properties, such as ultracold gases. While quantum computers can theoretically one day be programmed to solve any problem, quantum simulators are already used to explore specific problems.

LASERS

Highly sensitive laser beams are used for measuring the ions' properties, such as temperatures.

Trapped ion simulator

It may be possible to simulate interactions in quantum magnetism using trapped ions that mimic this magnetic behavior. This would not be feasible using a classical computer.

Ions are arranged in a matrix using magnetic fields and laser beams.

